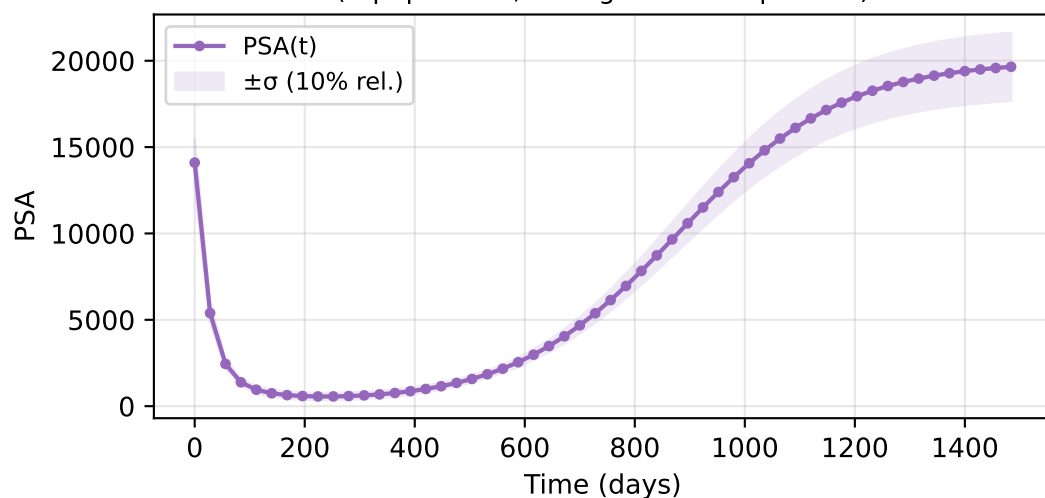
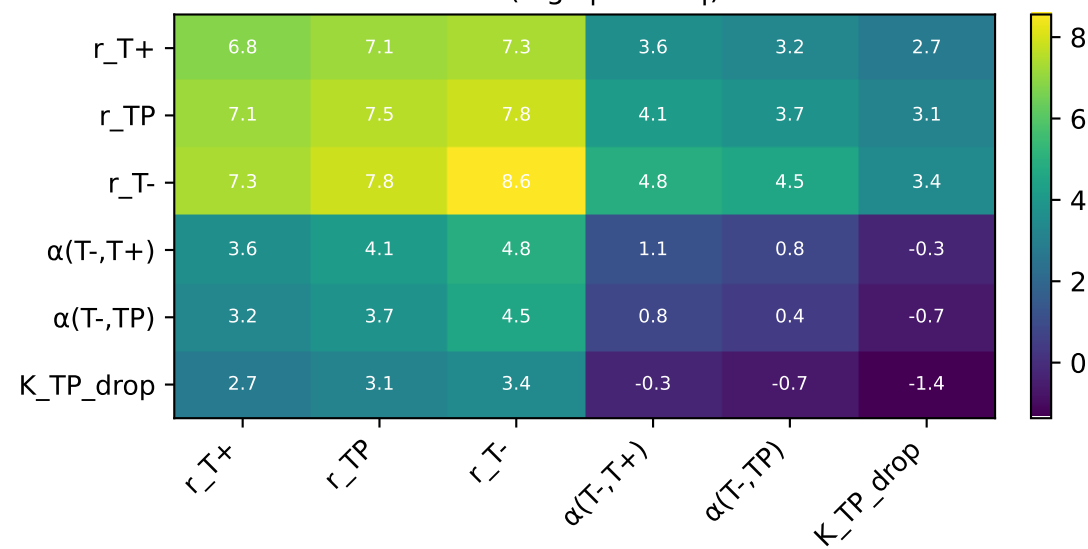


FIM identifiability — 3-pop K-shift Zhang model under MTD, PSA-only obs (N=54 samples)

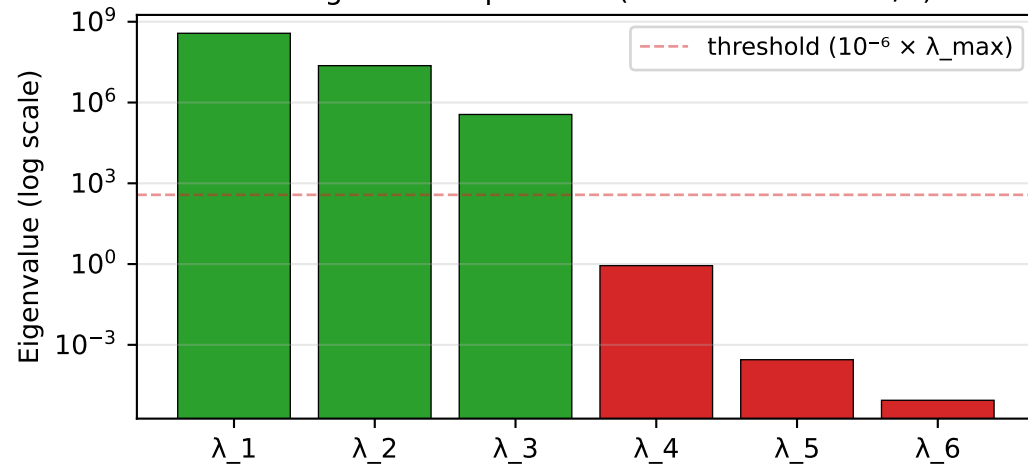
Nominal PSA trajectory under MTD
(3-pop K-shift, Zhang canonical params)



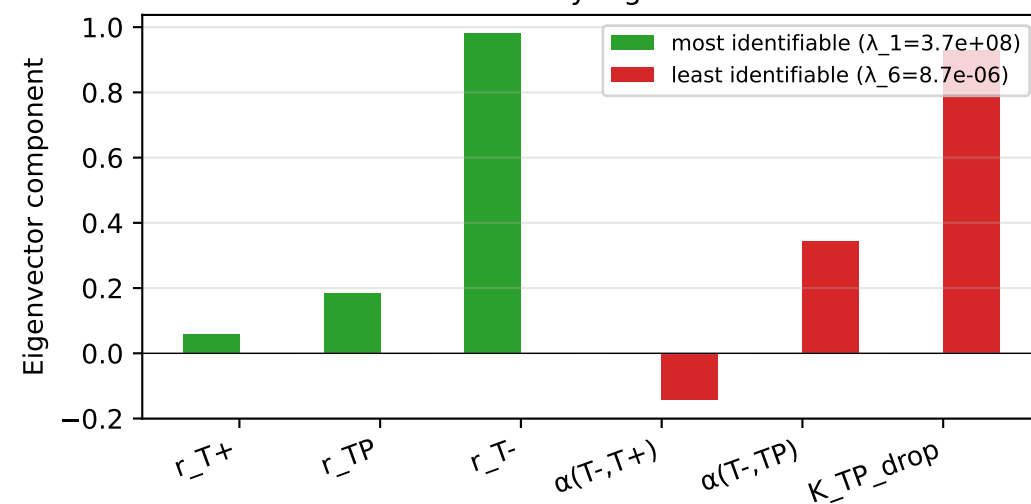
FIM ($\log_{10}|\text{entries}|$)



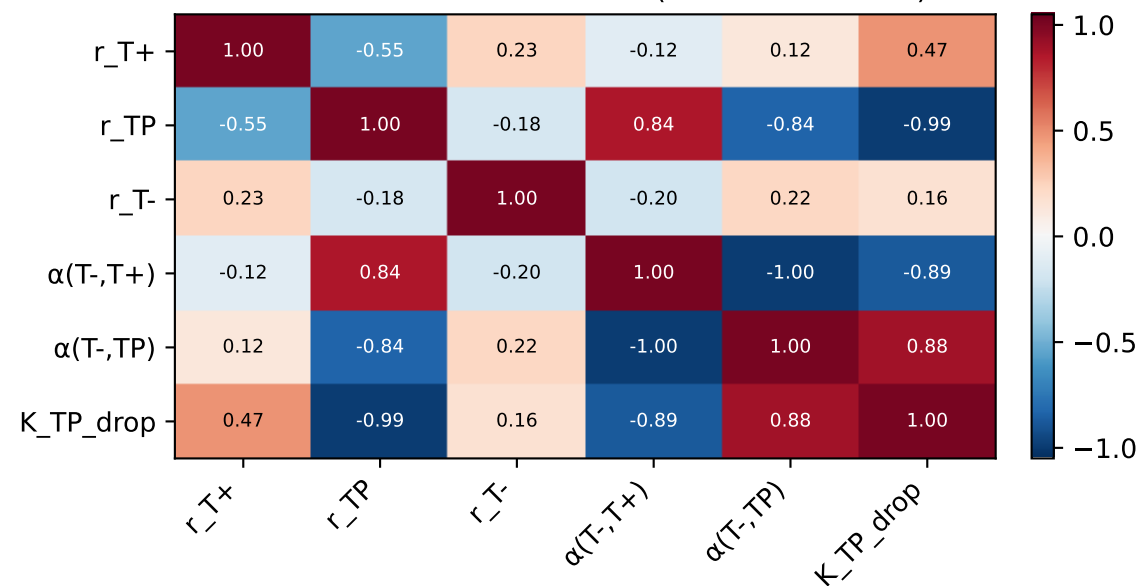
FIM eigenvalue spectrum (effective rank = 3/6)



Identifiability eigenvectors



Estimate-correlation matrix (FIM $^{-1}$ normalized)



Per-parameter sensitivity over time

